

# The Distributional Impacts of U.S. Transfer Programs\*

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## Abstract

Year-to-year income mobility dampens the ability of taxes to redistribute consumption. We show that eight U.S. transfer programs are nevertheless about as progressive with respect to consumption as they are with respect to income. The consumption-poorest people receive as much on average in transfers as the income-poorest, and transfers decline as quickly in consumption as in income. Why? Transfer take-up among eligible people is highly selected towards those with low consumption given income, offsetting mobility's effect, while eligibility rules select more weakly. Due to these selection forces, transfers give twice as much to the consumption poor as “distributionally equivalent” income taxes.

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# 1 Introduction

People move up and down the distribution of annual income throughout life, due to idiosyncratic shocks and life-cycle forces. This year-to-year income mobility reduces the power of progressive taxation of annual income to address inequality in consumption and lifetime income. As higher-income and lower-income people see their incomes revert towards the mean over time, tax burdens tend to be more uniform as shares of consumption (Poterba, 1989, 1991) and lifetime income (Fullerton and Lim Rogers, 1993) than they are as shares of annual income.

Is this also true of transfer programs? Many people with transiently low income can smooth consumption (Meyer and Sullivan, 2003, 2012; Sullivan et al., 2008). If these people are eligible for and take up transfer benefits amid these low-income spells, then transfer programs may redistribute less to the consumption poor than their progressivity with respect to income would suggest. Yet transfers could instead be as or more progressive in consumption than in income. Programs often define eligibility using attributes, like single parenthood or disability status, that are informative about consumption and lifetime income. Eligible households also choose whether to participate in transfer programs and may do so selectively according to their consumption or lifetime income, with that selection either towards the neediest (Nichols and Zeckhauser, 1982) or perversely toward those least in need (e.g., Deshpande and Li, 2019; Herd and Moynihan, 2019). Eligibility rules and selective take-up may therefore enhance or attenuate the power of transfers to redistribute consumption, offsetting or amplifying the effects of income mobility.

We show that eight U.S. transfer programs with a total budgetary cost in 2019 of about \$800 billion are, taken together, about as progressive with respect to consumption as they are with respect to income. Despite year-to-year income mobility, both receipt rates and average annual per-capita benefits are as high at the bottom of the distributions of consumption and lifetime income as at the bottom of the income distribution. In total across the eight programs from 1997 to 2017, the consumption-poorest and lifetime-poorest people received an annual per-capita transfer of approximately \$3,400 in constant 2020 dollars, relative to a transfer of \$3,600 to the current-poorest people. Receipt rates and average benefits also fall about as quickly with respect to consumption rank and lifetime-income rank as they do with respect to current-income rank. Overall, the U.S. social safety net does as much to redistribute to the consumption- and lifetime-poor as the current poor, quite unlike income taxation.

Transfers remain strongly progressive in consumption terms because they target people with low consumption given income. Mobility, eligibility, and take-up collectively determine how the benefits of transfer programs are targeted over the consumption and lifetime-income distributions relative to the current-income distribution. Focusing on the Supplemental Nutrition Assistance Program (SNAP), the average consumption rank of those enrolled is 15 percentiles lower than those not enrolled but with similar annual incomes. This result holds even when comparing transfer recipients to eligible non-recipients, pointing to the primacy of selection via take-up rather than selection via eligibility. We detect differences of similar, if typically slightly smaller, magnitude for other programs and for lifetime income. A set of demographic controls

can explain some but not all of the negative selection on consumption and lifetime income of transfer recipients. Take-up thus reflects private information about need as well as information observable to the government.

We introduce a decomposition that defines and measures the contributions of mobility, eligibility, and take-up. We find that, absent eligibility and take-up effects, average total per-capita benefits among the consumption-poorest and lifetime-poorest households would be around two-thirds of its actual level. Selective take-up appears more important than eligibility rules in concentrating the incidence of transfer programs at the bottom of the consumption and lifetime-income distributions. Furthermore, the mobility effect implies there would be, but for eligibility rules and selective take-up, substantial incidence of transfer programs at high percentiles of the consumption and lifetime-income distributions.

To be distributionally equivalent in consumption terms, income taxes must provide more to the current-poor and have higher marginal tax rates than transfers. The income tax that best replicates the transfer system's consumption incidence provides nearly \$6,000 to the bottom of the current-income distribution, compared with \$3,500 provided by the transfer system. The replicating income tax has an average marginal tax rate from the bottom to the tenth percentile of the income distribution of 25 percent. By contrast, the transfer system's average marginal tax rate from the bottom to the tenth percentile of the income distribution is 14 percent. Because of the targeting power of selective take-up and eligibility rules, transfers can provide less and have lower marginal tax rates at the bottom of the current-income distribution than the replicating income tax.

Our study uses Panel Study of Income Dynamics (PSID) data from 1997 to 2017. These data allow us to examine the distributional incidence of several transfer programs with respect to both consumption and lifetime income in a nationally-representative sample, an analysis that would be infeasible with existing U.S. administrative data. Administrative data typically lack consumption or long-run measures of income and are not comprehensive at the national level across transfer programs. The misreporting of benefit receipt in surveys poses, in our view, the most important challenge to our analysis (Meyer et al., 2009, 2015). In particular, a recent literature finds that misreporting probabilities vary with household characteristics (Davern et al., 2019; Mittag, 2019; Meyer et al., 2020). The same literature has, fortunately, also estimated multivariate corrections for misreporting that are intended to address this problem. We apply their corrections in the cases of SNAP and Medicaid and find they leave our conclusions essentially unchanged. The equal progressivity of transfers with respect to consumption and income is thus unlikely to be an artifact of underreporting of transfer receipt.

We contribute to a literature in public finance on the distributional incidence analysis of public policies, as last reviewed in Fullerton and Metcalf (2002). These analyses study taxes and transfers across the annual income distribution, and they often influence U.S. policymaking.<sup>1</sup>

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<sup>1</sup>The U.S. Congressional Budget Office has produced such an annual report, now titled "The Distribution of Household Income," since 2011. The Joint Committee on Taxation, the Urban-Brookings Tax Policy Center, and the Tax Foundation also produce, for some pending legislation, distributional analyses of the kind this paper considers.

Much economic research on this topic emerged out of a desire to understand the limitations of such analyses that had weighed on policy debates (e.g., Auerbach and Kotlikoff, 1987; Bradford, 1995). Following the classic work of Pechman and Okner (1974), these contributions mostly focused on taxes rather than transfers. Those which did study transfers often assumed that benefit receipt was a function of current income, ruling out the eligibility and take-up channels we explore in this paper (Davies et al., 1984; Fullerton and Lim Rogers, 1993). Other parallel research suggests the potential importance of these channels to distributional analysis: Scholars have found heterogeneous income dynamics among low-income households (Bane and Ellwood, 1986) and selective take-up among the eligible (Blank and Ruggles, 1996; Currie, 2006; Bhargava and Manoli, 2015; Finkelstein and Notowidigdo, 2019). We show that, for most U.S. transfer programs, the eligibility and take-up channels are so important as to fully offset the effects of mobility on incidence. Our results therefore lend support to theories like Nichols and Zeckhauser (1982) in which incomplete take-up improves targeting.

Our paper is, to the best of our knowledge, the first systematic study of the consumption incidence of U.S. transfer programs. It is most closely related to several recent papers that consider the distributional incidence of taxes and transfers from a lifetime perspective.<sup>2</sup> Roantree and Shaw (2018) and Levell et al. (2021) study the current versus lifetime redistribution of specific transfer programs in the United Kingdom. Brewer et al. (2020) proposes a related decomposition, delineating how policy reforms affect the lifetime-income distribution due to static impacts, income dynamics, and tagging of individuals with heterogeneous income dynamics.

We extend their contributions in three main ways. First, we use data on transfer receipt rather than predicted receipt from microsimulation models. This distinction is essential in the context of distributional analysis, as microsimulation models impose assumptions about the relationship between transfer receipt, consumption, and income. This paper relaxes these assumptions and shows that transfer receipt predicts consumption and lifetime income conditional on eligibility and current income. Simulating receipt can therefore portray transfer programs as less progressive than they actually are. Second, we link research on the lifetime incidence of transfer programs to the incomplete take-up literature. Relative to other developed countries, incomplete take-up is particularly relevant in the U.S., where many social programs are not universal entitlements. Moreover, with data on both transfer eligibility as well as receipt, we consider whether the “tagging effect” of Brewer et al. (2020) occurs because households are ineligible for transfers or because they choose not to take them — a core distinction in welfare analysis as well as in policymaking. Third, we examine the distributional incidence of transfers with respect to consumption and lifetime income. Consumption data provide a transparent benchmark for estimated lifetime incomes from lifecycle models, which inevitably require significant assumptions to address limitations of existing household panel data. Furthermore, in the context of transfer programs, consumption may be a superior proxy for the marginal utility of income, due to

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<sup>2</sup>Bengtsson et al. (2016) and Auerbach et al. (2021) also assess the distributional incidence of the Swedish and American consolidated tax-and-transfer systems.

borrowing constraints among transfer recipients.

This paper proceeds as follows. Section 2 introduces the data. Section 3 presents the central descriptive facts of this paper, that most transfer programs are about as progressive with respect to consumption and lifetime income as they are with respect to income. Section 4 then explains how the negative selection on consumption and lifetime income conditional on current income yields this fact, and assesses the relevance of eligibility rules and selective take-up in producing this selection pattern. Section 5 extends our approach to show how much more effective transfer programs are in reducing consumption poverty than a negative income tax with the same distributional incidence with respect to income. Section 6 concludes.

## 2 Data and Measurement

Our main source of data is the Panel Study of Income Dynamics (PSID) in its eleven biennial survey waves from 1997 to 2019.<sup>3</sup> In each PSID wave, we observe heads of household and spouses from age 18 to 65. We briefly review key aspects of the data and provide further details in Appendix B. Two limitations of the PSID data—namely, the absence of cash-equivalent values of in-kind transfers and measures of transfer eligibility—require us to draw on additional sources of data also described here.

The PSID is the only U.S. dataset suitable for our analysis. It has a uniquely long panel dimension relative to other surveys. Administrative data do have some advantages, discussed below, but they lack core elements required for our analysis. Most importantly, transfer records are often housed at disparate state agencies and are infeasible to link across all states and programs. Administrative data also do not contain measures of consumption.

**Current Income.** We define household income as total annual income of the head and spouse before taxes and transfers, excluding other household members due to data constraints. Income includes labor, business, and capital income. Following the National Academy of Sciences (Citro and Michael, 1995), we adjust for household size using the equivalence scale  $e_h = (N_{h,adult} + 0.7N_{h,child})^{-0.7}$ , where  $N_{h,adult}$  and  $N_{h,child}$  respectively denote the numbers of adults and of children in household  $h$ .<sup>4</sup>

**Current Consumption.** The PSID has included extensive coverage of consumption expenditures since 1999. Expenditure categories covered include food, housing, health care, transportation, education, child care, and several smaller topics. We make two notable adjustments to the data to better measure consumption rather than expenditure. First, for homeowners, we replace mortgage and property tax payments with equivalent rents based on reported home values. Second, for vehicle owners, we replace loan payments with the annual depreciation based on reported

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<sup>3</sup>Several key variables for our analysis are not available in PSID waves before 1997.

<sup>4</sup>Appendix A includes results not adjusted for household size and composition. These are typically quite similar to our equivalized-household results. One implication is that results using other equivalence scales are likely to be almost indistinguishable from our main results.

vehicle values. Household consumption is then equivalized by the same scale as income.

**Transfer Receipt.** We observe self-reported household receipt for nine transfer programs. These are the Supplemental Assistance Nutrition Program (SNAP); Medicaid; Section 8; public housing; Temporary Assistance for Needy Families (TANF); Supplemental Security Income (SSI); unemployment insurance (UI); Women, Infants, and Children (WIC); and the Low Income Home Energy Assistance Program (LIHEAP). In the paper, we combine public housing and Section 8 into a single program that we call “housing assistance,” but we present results that distinguish between them in Appendix A.

**Valuing Transfer Benefits.** We measure the dollar value of transfer benefits by combining information from the PSID and the Supplemental Poverty Measure module of the U.S. Current Population Survey (CPS). For SNAP, TANF, SSI, UI, and LIHEAP, the PSID records the nominal value of benefits over various time periods, which we rebase as the per-capita annualized amount in 2020 constant dollars. The PSID does not include cash-equivalent values for in-kind transfer benefits, namely Medicaid, Section 8, public housing, and WIC. We impute these amounts with the average values by household size and year reported in the CPS data.

**Lifetime Income.** We construct a lifetime concept of household income from incomplete income histories. To begin, we estimate Poisson regression models with individual fixed effects—the textbook “errors-in-variables” model for lifetime income and address Haider and Solon (2006)’s criticism of this model in Appendix B. Letting  $i$  index individuals,  $t$  index calendar years, and  $a$  index age in years, we estimate models of the following form:

$$E[y_{it} | \mathbf{X}_{it}] = \exp(\alpha_i + \alpha_t + \mathbf{X}_{it}'\boldsymbol{\beta}_a), \quad (1)$$

where  $\alpha_i$  is an individual fixed effect,  $\alpha_t$  is a year fixed effect,  $\mathbf{X}_{it}$  is a matrix of time-varying demographic characteristics, and  $\boldsymbol{\beta}_a$  is a vector of coefficients which may be interacted with age. The outcome  $y_{it}$  is individual income.

We then perform three adjustments before using the regression results to estimate lifetime income. First, we apply empirical Bayes methods from Morris (1983) to shrink the estimates  $\hat{\alpha}_i$  of the individual fixed effects towards a conditional expectation fit from time-invariant individual characteristics. Second, we balance the panel, imputing demographic characteristics as necessary, and calculate lifetime average income from ages 18–65. Third, we account for spousal income in a way that permits changes in household membership. If we were to restrict our sample to stable households (as in, e.g., Fullerton and Lim Rogers, 1993), our definition of their lifetime income would coincide with the natural concept. Let  $j(i, t)$  indicate  $i$ ’s spouse in year  $t$ . Our concept of lifetime household income follows each individual through the sequence of households they experience as adults. That is, the lifetime household income of individual  $i$  is

$$\bar{y}_i^h = \sum_t e_h \hat{y}_{it}^h = \sum_t e_{h,t} (\hat{y}_{it} + \hat{y}_{j(i,t),t})$$

where  $t$  is again summed over the years in which  $i$  is between ages 18 and 65, and  $e_{h,t}$  is the equivalence-scale adjustment for household  $h$  in year  $t$ .

**Simulated Eligibility.** We simulate transfer eligibility in the PSID by compiling information on program rules, mainly from primary-source documents and research databases of such rules, similar to the Urban Institute’s TRIM program (Zedlewski and Giannarelli, 2015). Our use of simulated eligibility is somewhat limited: to assess the likely importance of eligibility rules versus selective take-up among the eligible (similar to, e.g., Blank and Card, 1991) as mechanisms for our central findings about distributional incidence.

Simulated-eligibility measures cannot perfectly measure true eligibility, as there are many differences between information used in actual eligibility determinations versus survey data. In validation checks in Appendix B, we confirm that our simulated eligibility measure is highly predictive of benefit receipt, though misclassification is apparent. Significant fractions of recipients are simulated to be ineligible, and take-up rates are counterfactually low among those simulated to be eligible. The main threat from mismeasured eligibility in our analysis is that we may understate the importance of eligibility rules in determining the distributional incidence of transfers. In robustness checks throughout the paper, we show how our results change when we reclassify simulated-ineligible transfer recipients as eligible, which mechanically raises the influence of eligibility rules relative to take-up.

### 3 The Progressivity of Transfer Programs

This section presents the two core descriptive facts of the paper. First, for most transfer programs, the receipt rate and the average dollar value of benefits per person per year about as high at the bottom of the consumption and lifetime-income distributions as they are at the bottom of the current-income distribution. Second, receipt rates and average benefits decline about as steeply in consumption and lifetime-income ranks as they do in current-income rank.

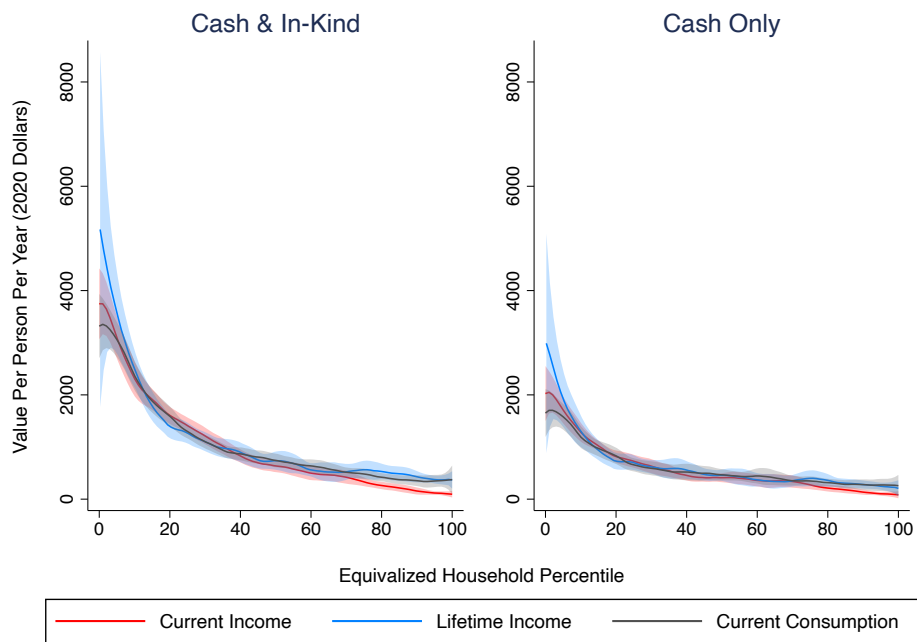
**Approach.** We estimate receipt rates and average benefits as locally-linear functions of households’ percentile rank in a given distribution. We rank households on the basis of their equivalized current income, equivalized current consumption, and equivalized lifetime income. Appendix A contains a wealth of supplementary figures that probe the sensitivity of our results to definitional choices and provide detail on distributional incidence.

**Average Total Transfer Per Person Per Year.** Panel A of Figure 1 plots the average total annual per-capita value of transfer benefits as functions of consumption rank, current-income rank, and lifetime-income rank. In the left sub-panel, we combine transfers across the eight transfer programs we study. The right sub-panel excludes in-kind programs, for which we must impute cash-equivalent values of benefits.

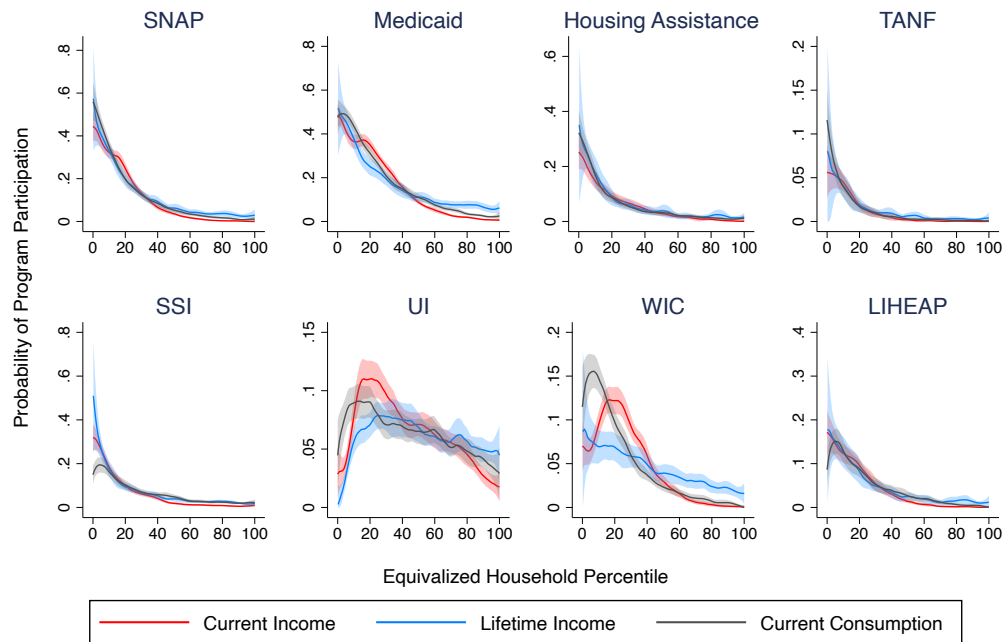
The figure shows that, overall, the U.S. social safety net is about as targeted at households with the lowest consumption and lifetime incomes as it is at those with the lowest current in-

Figure 1: Receipt and Value of Transfer Benefits as a Function of Equivalized Household Rank

Panel A: Average Total Annual Per-Capita Benefits



Panel B: Receipt Rate by Program



*Notes:* This figure displays the average total annual per-capita value of transfer benefits and receipt rates by program as functions of household ranks in the distributions of equivalized current income, lifetime income, and current consumption. The functions are estimated by local linear regressions with bandwidths of three percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands.



comes. Average annual per-capita benefits are about as high at the bottom of the distributions of consumption and lifetime income as they are at the bottom of the current-income distribution. The similarity of the results for consumption and lifetime income suggests transfer programs identify households with low consumption as a result of persistently low income, rather than a lesser ability to smooth consumption relative to income over time. Transfer payments also decline as fast in consumption rank and lifetime-income rank than as current-income rank. We construct consumption and lifetime-income ranks separately and thus view the similarity of these results as reassuring for our estimates of lifetime income.<sup>5</sup>

**Receipt Rates and Amounts by Program.** Panel B of Figure 1 plots estimates of transfer receipt rates by consumption rank, current-income rank, and lifetime-income rank for the eight transfer programs we study. Receipt rates are generally about as high at the bottom of all three distributions, without the clear pattern of more-uniform consumption incidence and lifetime incidence that year-to-year income mobility would suggest.

The programs that are most additionally progressive in consumption terms relative to their current incidence are SNAP, Medicaid, and housing assistance programs. Each has receipt rates that are roughly 20 percentage points higher at the bottom of the consumption and lifetime-income distributions than at the bottom of the current-income distribution. SSI is particularly generous to the lifetime-poorest, but not necessarily to the consumption-poorest. This combination of programs already suggests that both eligibility rules and selective take-up are both important contributors to the lifetime progressivity of transfer programs. Among working-age adults, SSI eligibility is tightly restricted to people with disabilities and trivial income and assets, whereas eligibility for SNAP and housing assistance are relatively broad, but take-up is highly incomplete and the programs feature notable ordeals.

**Adjustments for Misreporting.** How sensitive are our results likely to be to misreporting of transfer receipt? Using linked survey and administrative data, Mittag (2019) and Davern et al. (2019) estimate statistical models of household survey reporting behavior for SNAP and Medicaid respectively. Their models, intended for use as misreporting corrections, predict the probability of true transfer receipt given survey-reported receipt and demographic characteristics. Intuitively, these models allow researchers to replace assumptions of constant misreporting rates with misreporting probabilities that are functions of demographic observables.

These corrections have little impact on our findings. We remake Panel B of our Figure 1 for SNAP and Medicaid in Appendix Figure A5. We first adjust results by assuming constant rates of false-positive and false-negative reports as in Meyer et al. (2009), which we show in the left panels. We calibrate these misreporting rates so that they match the overall misreporting rates implied by the Davern et al. (2019) and Mittag (2019) models in our data. Then, in the right panels, we directly apply the multivariate misreporting corrections in these papers. For both Medicaid and SNAP, receipt rates at the bottom of the distributions of income, consumption,

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<sup>5</sup>Caspersen and Metcalf (1994), one of the few other studies to analyze both consumption incidence and lifetime incidence, also find that they are quite similar to each other.

and lifetime income all remain approximately the same after applying either correction.

## 4 Why Are Transfers So Progressive in Consumption?

Transfer programs are notably progressive with respect to consumption and lifetime income, relative to their incidence with respect to income. Why? This section shows that, across most major U.S. transfer programs, recipients have sharply lower consumption and lifetime incomes than non-recipients with the same current incomes. This negative selection on consumption and lifetime income offsets year-to-year income mobility, which would otherwise make incidence with respect to consumption and lifetime income more uniform than current-income incidence.

We first document the negative conditional selection of transfer recipients on consumption and lifetime income. Second, we use our simulated-eligibility data to assess the likely importance of eligibility rules and selective take-up among eligible people in creating this pattern of selection. Third, we apply a formal decomposition of mobility, eligibility, and take-up effects on distributional incidence that is related to Brewer et al. (2020).

### 4.1 Selection into Transfer Receipt

Transfer recipients have lower consumption and lifetime income than non-recipients with the same current income. We document this fact in rank–rank regressions. We then show it holds even after controlling for simulated eligibility and, to a lesser extent, controlling for an array of observable characteristics. Our primary regression specification is

$$\bar{R}_{it} = \alpha_{ct} + \beta R_{it} + \gamma D_{it} (+\Omega_{it}) + u_{it}, \quad (2)$$

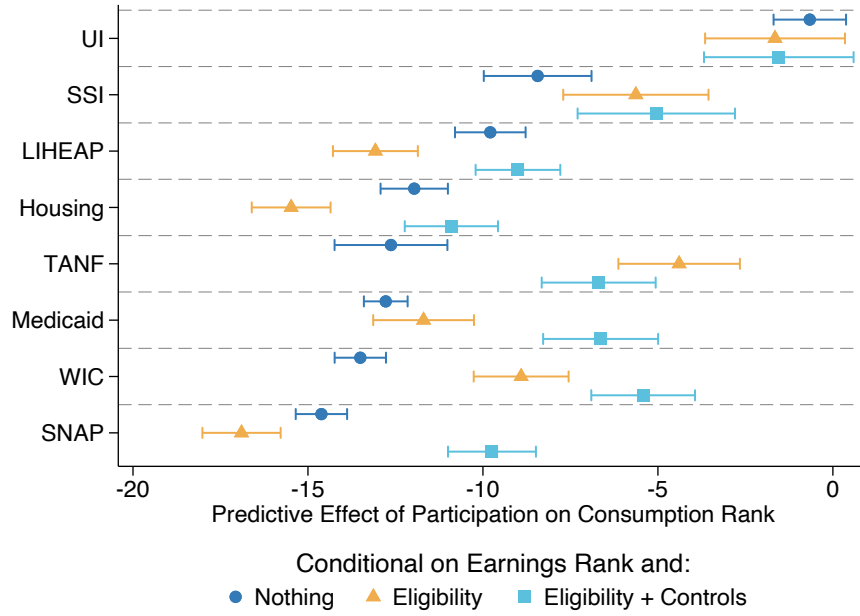
where  $\bar{R}_{it}$  is the consumption rank or lifetime-income rank for person  $i$  in year  $t$ ,  $R_{it}$  is  $i$ 's current-income rank,  $D_{it}$  indicates  $i$ 's receipt status for a given transfer program, and  $\alpha_{ct}$  is a cohort–year fixed effect. In some specifications, we restrict the sample to people who are simulated-eligible and include demographic controls  $\Omega_{it}$ .

Figure 2 shows that transfer receipt is associated with a lower consumption rank and lifetime-income rank, conditional on current-income rank (blue circles). Across programs, transfer recipients rank on average 10 to 15 percentiles lower in the consumption distribution than non-recipients of that program with the same current-income rank. The extent of conditional selection into transfer receipt on consumption and lifetime rank varies considerably across transfer programs: SNAP and Medicaid are highly informative about consumption and lifetime income, whereas UI is essentially uninformative about consumption given income. In a few cases there are marked differences between selection on consumption and selection on lifetime income. Most notably, SSI recipients are highly negatively selected on lifetime income given current income but are much less selected on consumption given current income.

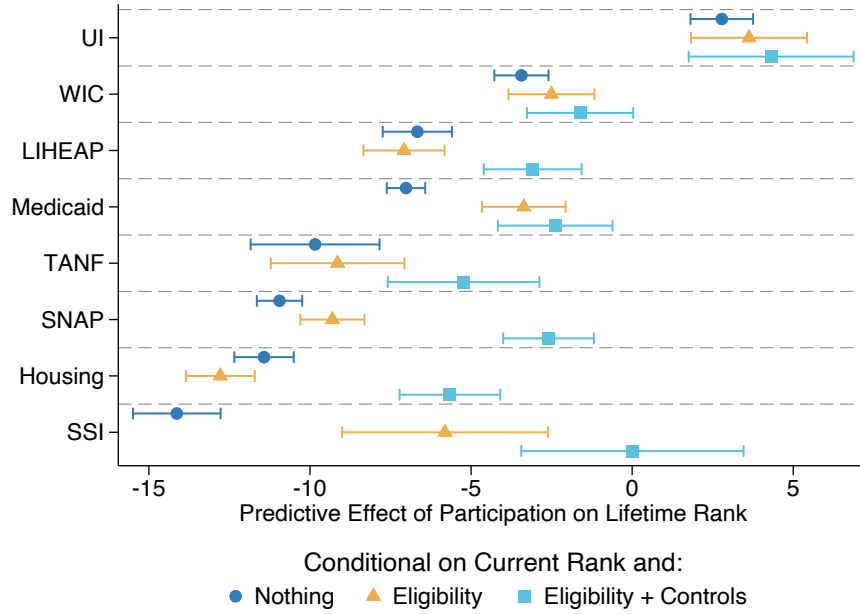
Conditioning on simulated eligibility does little, on average, to reduce the average gaps in

Figure 2: What Explains Selection into Transfer Receipt?

Panel A: Selection on Consumption



Panel B: Selection on Lifetime Income



*Notes:* This figure displays estimates of the predictive effect of transfer receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient  $\gamma$  from Equation 2). For estimates represented by blue circles, we do not add additional control variables to the specification, whereas for the yellow diamonds, we estimate the regression only on people whom we simulate to be eligible, and for the teal squares, we subset to the simulated-eligible population and control for a rich set of demographic variables. The confidence intervals are for the 95-percent level and reflect clustered standard errors by individual.

consumption and lifetime income between recipients and non-recipients (yellow triangles). Most of the selection into transfer receipt on low consumption given income therefore reflects selective take-up among the eligible, rather than eligibility rules that screen out people who can smooth consumption through low-income spells. This conclusion largely does not appear to be the result of noisy measures of eligibility. When we reclassify any simulated-ineligible recipients of a given transfer program to be eligible for that program, we do not see a clearer role for eligibility in concentrating incidence among the consumption poor (Appendix Figure A7). The transfer program where our results are potentially most sensitive to mismeasurement of eligibility is SSI. Even with this adjustment to our eligibility measure, take-up remains the primary force in shaping the consumption incidence of transfers.

Is selective take-up among the eligible best viewed as selection on private information or just the “incompleteness” of existing eligibility rules? We re-estimate Equation 2 on the simulated-eligible population and include a set of demographic characteristics that are observable in principle to the government (teal squares).<sup>6</sup> Adding these controls eliminates some but not all of the remaining selection into transfer receipt for most programs. While some of the informational content of take-up reflects observables that could be further used in eligibility rules, it does contain a component of private information about need. This analysis could also be interpreted as further evidence that, even if we had the ideal measure of eligibility, it would not overturn a substantial role for take-up. Among demographically-similar people, and thus people with likely-equivalent true eligibility, transfer receipt remains highly informative of consumption and lifetime income.

**Measurement Error.** We conduct adversarial simulations that assume various rates of under-reporting of transfer receipt at the top of the consumption distribution (see Appendix B). Our conclusions are robust even with a degree of underreporting that we view as implausible, such as “false-negative” rates of 50 percent in the top quarter of the consumption distribution.

## 4.2 A Decomposition of Distributional Incidence

We have so far argued informally through regressions that take-up, more so than eligibility, explains why transfer receipt identifies those with low consumption given income. We now develop a decomposition, related to Brewer et al. (2020), which more carefully distinguishes between selection via eligibility rules and selective take-up among the eligible.

**Mobility Effect.** The mobility effect is the component of the difference between incidence with respect to consumption and incidence with respect to income that results from year-to-year income mobility. For instance, because college graduates have significant debt early in their careers but later greatly out-earn non-graduates, the lifetime incidence of student loan forgiveness is less

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<sup>6</sup>We condition on covariates that are used in eligibility rules for at least one transfer program. These are the household’s state of residence, household size and composition, family income and earnings, ages of household members, disability status, unemployment duration and reason, and basic measures of wealth (value of any automobiles and liquid assets).

progressive than its incidence with respect to current income (Catherine and Yannelis, 2020).

To measure the mobility effect, we estimate the share  $q(r)$  of people with consumption rank  $\bar{R}_i = r$  who would have received a given transfer in a given year if, counterfactually, transfer receipt were only a function of current-income rank. That is, our mobility-only counterfactual is

$$q(r) = \int \Pr(D_{it} = 1 \mid R_{it}) dF(R_{it} \mid \bar{R}_i = r),$$

where  $D_{it}$  indicates transfer receipt and  $F(R_{it} \mid \bar{R}_i = r)$  is the conditional distribution of current-income ranks  $R_{it}$  at consumption rank  $r$ . The mobility effect at rank  $r$  is the difference between  $q(r)$  and the empirical receipt rate at a current-income rank  $r$ ,  $p(r) = \Pr(D_{it} = 1 \mid R_{it} = 1)$ . Intuitively, the mobility effect applies the consumption–income “transition matrix” to the receipt rate by income rank, yielding the counterfactual receipt rates at each consumption rank.

**Eligibility Effect.** The eligibility effect is the component of the difference in incidence due to eligibility rules that, among low-income households, tag those with low consumption and low lifetime income. For instance, asset tests and categorical eligibility for single parents, people with disabilities, and similar groups tend to target benefits to the persistently poor, not people with merely low current income. By contrast, eligibility rules for contributory social insurance programs such as unemployment insurance operate in the opposite direction, restricting transfers to the persistently poor.

To measure the eligibility effect, we define the receipt rate at consumption rank  $r$  under a second counterfactual,  $s(r)$ . Here the probability of take-up conditional on eligibility is a function of current income alone, whereas we let eligibility depend on both current income and consumption (or current income and lifetime income). This counterfactual receipt rate  $s(r)$  thus depends only on eligibility and current-income rank. The difference between these two counterfactuals,  $q(r)$  and  $s(r)$ , is the eligibility effect.

When eligibility can be measured perfectly—that is,  $E_i = 0$  implies  $D_i = 0$ —our second counterfactual is defined as

$$s(r) = \int \Pr(D_{it} = 1 \mid R_{it}, E_{it} = 1) \Pr(E_{it} = 1 \mid R_{it}, \bar{R}_i = r) dF(R_{it} \mid \bar{R}_i = r),$$

where  $E_{it}$  indicates eligibility. This expression emerges from the following reasoning. The probability  $\Pr(D_{it} = 1 \mid R_{it}, E_{it} = 1)$  is the transfer take-up rate among the eligible at current-income rank  $R_{it}$ . The probability  $\Pr(E_{it} = 1 \mid R_{it}, \bar{R}_i = r)$  is the eligibility rate at current-income rank  $R_{it}$  and consumption rank  $r$ . Multiplying these two probabilities yields a predicted receipt rate for people with current-income rank  $R_{it}$  that embeds the desired independence assumption: Given current income, take-up among the eligible is uninformative about consumption or lifetime income.<sup>7</sup> Integrating over the conditional distribution of current-income ranks given consumption,

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<sup>7</sup>To better understand this counterfactual, note that the law of conditional probability implies that:  $\Pr(D_{it} = 1 \mid \bar{R}_i = r) = \int \Pr(D_{it} = 1 \mid R_{it}, \bar{R}_i = r) dF(R_{it} \mid \bar{R}_i = r)$ . By consequence, the counterfactual receipt rate  $s(r)$  equals the empirical receipt rate at a given consumption rank if and only if take-up among eligibles is conditionally independent

$F(R_{it} | \bar{R}_i = r)$ , we have a counterfactual receipt rate  $s(r)$  at consumption rank  $r$  that permits a role for eligibility rules while shutting down selective take-up.

Imperfect measurement of eligibility requires a more complex expression for the same counterfactual concept, which we present in Appendix B.

**Take-Up Effect.** The take-up effect is the component of the difference in incidence that results from the influence of consumption and lifetime income on take-up rates among the eligible. For instance, procedural complexity (Deshpande and Li, 2019; Gray, 2019), “ordeal” costs (Nichols and Zeckhauser, 1982), and information frictions (Finkelstein and Notowidigdo, 2019) could select for or against low lifetime income and consumption among eligible households with similar current income. We define the take-up effect as the residual difference between the counterfactual receipt rate  $s(r)$  and the empirical receipt rate at the same consumption rank,  $\bar{p}(r) = \Pr(D_{it} = 1 | \bar{R}_i = 1)$ .

### 4.3 Results of Decomposition

This decomposition enables us to interpret differences between the consumption incidence and the income incidence of transfer programs. For instance, why do a greater or smaller share of people at consumption rank  $r$  receive a transfer than do people at current-income rank  $r$ ? Our decomposition is

$$\bar{p}(r) - p(r) = \underbrace{[\bar{p}(r) - s(r)]}_{\text{take-up effect}} + \underbrace{[s(r) - q(r)]}_{\text{eligibility effect}} + \underbrace{[q(r) - p(r)]}_{\text{mobility effect}}, \quad (3)$$

using the objects defined above. Figure 3 displays decomposition results for the difference between current-income and consumption incidence. Panel A displays results for the average total annual per-capita value of benefits, and Panel B displays results for receipt rates by program.<sup>8</sup> The blue, black, and yellow lines plot the mobility, eligibility, and take-up effects, respectively; each represents the difference in consumption incidence with respect to current-income incidence if only this effect were present. The blue shaded region indicates the net effect, equal to the difference between the consumption and current-income incidence at the indicated rank; this region integrates to zero.

Absent eligibility and take-up effects, the incidence of transfer programs would be markedly less progressive at the bottom of the distribution. Consistent with our prior results, we find a central role for selective take-up among eligible households, and a modest role for eligibility rules, in shaping the consumption incidence of transfer programs. UI is the main exception to our results, with little role for either eligibility or take-up effects.

In total over transfer programs, selective take-up increases the average annual value of benefits received by the consumption-poorest people by about \$750 per person relative to a counterfac-

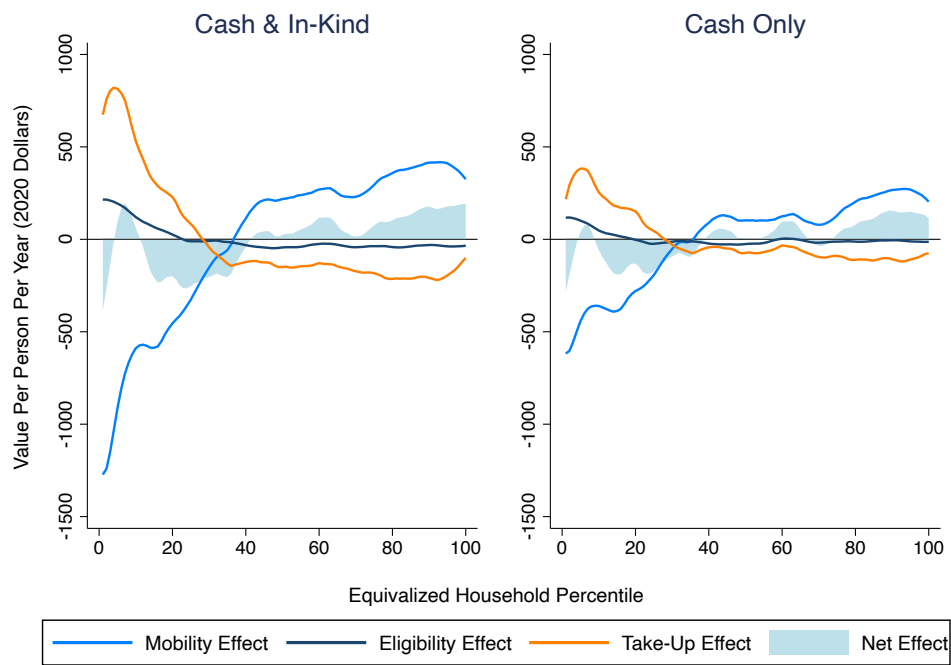
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of consumption given current income at all consumption ranks.

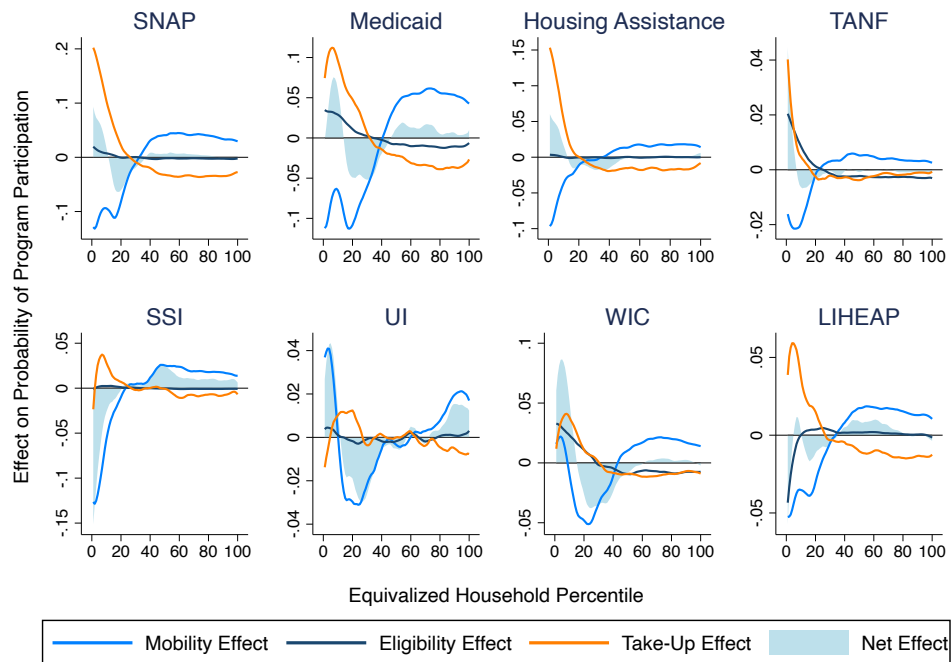
<sup>8</sup>Appendix A contains supplementary figures that step through the decomposition and probe sensitivity to definitional choices.

Figure 3: Decomposition of Difference Between Current-Income and Consumption Incidence

Panel A: Average Total Annual Per-Capita Benefits



Panel B: Receipt Rate by Program



*Notes:* This figure displays decompositions of differences between current-income and consumption incidence into mobility, eligibility, and take-up effects. The underlying functions are estimated using local linear regressions with bandwidths of three percentiles.

tual in which transfer receipt is a function of current income and eligibility alone. Eligibility rules contribute about \$250 per person at the bottom of the consumption distribution. Together these amounts represent about one quarter of the average transfer per capita per year to consumption-poorest people. Mobility tends to shift the incidence of transfers up the consumption distribution, but for most programs, the eligibility and take-up effects largely offset the mobility effect.

The primacy of take-up applies broadly across the eight transfer programs we study. Eligibility rules appear most important in Medicaid, TANF, and WIC, whereas they appear essentially irrelevant or operating in the opposite direction for SNAP, LIHEAP, and housing assistance. While mobility alone would reduce the receipt rate among the consumption poor by about 10 percentage points for SNAP, Medicaid, and housing assistance, selective take-up increases receipt rates in these programs at the bottom of the consumption distribution by 10 to 20 percentage points. These are economically large differences relative to underlying rates of program participation as well as relative to the effects of mobility. In the case of UI, neither eligibility nor take-up is important in shaping incidence.

We test the sensitivity of these results to mismeasurement of eligibility in Appendix Figure A8. Reclassifying simulated-ineligible transfer recipients to be eligible, we find that eligibility rules become somewhat more important for Medicaid and WIC, though take-up remains important for these programs. Our results are, again, most sensitive for SSI. We view our results as indeterminate about the relative importance of eligibility versus take-up in the case of SSI. Even allowing for substantial mismeasurement of eligibility, take-up appears important, especially for SNAP, LIHEAP, and housing assistance.

## 5 Taxes Versus Transfers

This section estimates the change in income taxes that would be necessary to replicate the consumption incidence of various transfer programs.

**Approach.** Imagine the government were to eliminate a transfer program and replace it with an income-tax reform. We want to find the reform that minimizes the net change in consumption incidence. Due to the eligibility and take-up effects of transfer programs, the “distributionally equivalent” tax reform will differ from the observed current-income incidence of the transfer program.<sup>9</sup> If and only if the net effect of eligibility and take-up is zero throughout the consumption distribution—and thus mobility alone shapes the consumption incidence of a transfer—then any taxes and transfers with the same income incidence will also have the same consumption incidence.

Let  $S(\tilde{r})$  be the average annual dollar value of transfers at consumption rank  $\tilde{r}$ , and let  $T(r)$  be the average change in income taxes at income rank  $r$ . The tax change  $T^*(r)$  that would exactly

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<sup>9</sup>We consider static effects, following the Envelope Theorem (Bradford, 1995).



offset the transfer cut throughout the consumption distribution is:

$$T^*(r) = \left\{ T(r) : \int T(r) dF(r|\tilde{r}) = S(\tilde{r}) \text{ for all } \tilde{r} \right\}, \quad (4)$$

where  $F(r|\tilde{r})$  is the conditional distribution of consumption ranks  $\tilde{r}$  at income rank  $r$ . This equation thus implicitly defines the income tax  $T^*(r)$  as that which, given the joint distribution of income and consumption, exactly replicates the consumption incidence  $S(\tilde{r})$  of a transfer.

Estimation of the replicating tax change  $T^*(\tilde{r})$  by Equation 4 is an ill-posed inverse problem (Horowitz, 2014), in the sense that  $T^*(\tilde{r})$  is highly sensitive to small changes in  $S(r)$ , as is common in estimation problems of this form. Following the statistics literature, we use Tikhonov regularization (generalized ridge regression) to reduce this sensitivity. In particular, we penalize the absolute value of the second derivative of  $T^*(r)$  according to a parameter  $\lambda$ :

$$T^*(r) = \arg \min_{T(r)} \int \left\| S(\tilde{r}) - \int T(r) dF(r|\tilde{r}) \right\| d\tilde{r} + \lambda \|T''(r)\|, \quad (5)$$

where  $\|\cdot\|$  indicates the  $L^2$  norm. This approach smooths fluctuations in the replicating tax change without introducing a bias to its level or slope. In implementation, we discretize the distribution of income and consumption ranks into percentiles, and we select the penalty  $\lambda$  on the numerical second derivative via cross-validation (see Appendix B for further details).

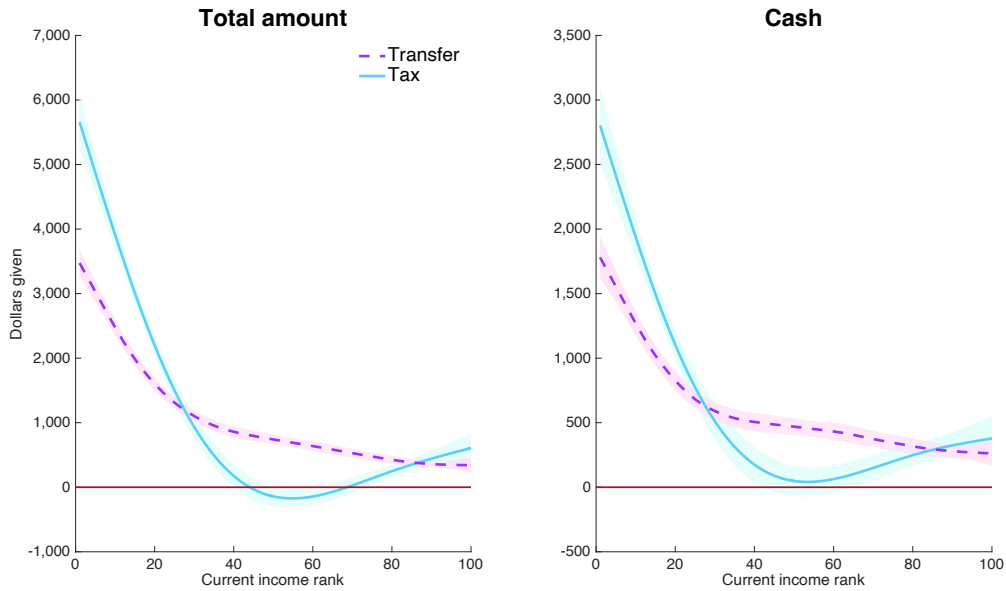
**Results.** We find that, due to the eligibility and take-up effects, the income taxes needed to replicate the consumption incidence of transfer programs are substantially more progressive in income terms than the transfer programs themselves. Figure 4 depicts, in purple lines, the average annual per-capita transfer payments over the current-income distribution, while the blue lines plot the change in income taxes needed to offset the transfer program in consumption terms. For comparability, we smooth these lines using the same regularization parameter.

To achieve the same consumption incidence as transfers, the income tax system would have to provide \$5,500 per person per year on average to those at the bottom of the current-income distribution, or \$2,000 more than the average transfer payment to this population. We easily reject that the replicating income taxes would be as progressive than the transfer system as a whole. Looking by program, the taxes needed to replicate each transfer program we study would have to be substantially more progressive, with the exceptions of UI and LIHEAP. For instance, the replicating taxes for SNAP would have to be about \$500 larger at the bottom of the income distribution than SNAP itself, or about two-thirds of the average SNAP payment to this population. This result reflects highly selective SNAP take-up among people with low consumption (Figure 3).

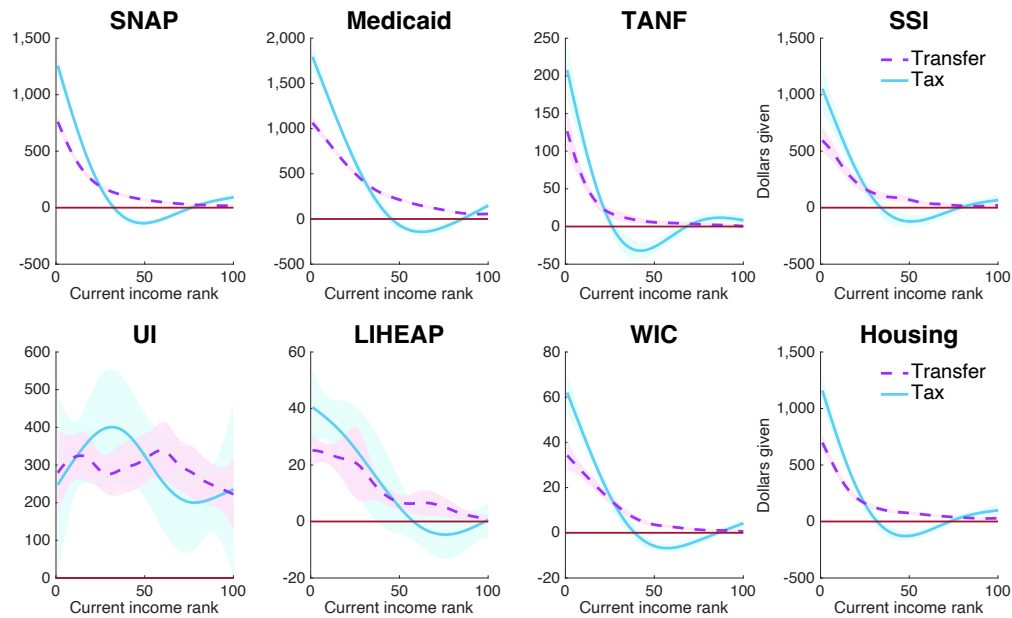
Appendix Table A2 presents the implied marginal tax rates at selected percentiles of the current-income distribution and average marginal tax rates up to that percentile. Marginal tax rates for the replicating tax are typically about twice as that of the transfer system. For example, the average marginal tax rate between from the bottom to the tenth percentile of the income

Figure 4: Using Income Taxes to Replicate Transfers

Panel A: Average Total Annual Per-Capita Transfer



Panel B: Amount by Program



*Notes:* This figure displays the average annual transfer payments received by people across the current-income distribution (purple line) and the negative income tax that would replicate the consumption incidence of the transfer program (blue line). Panel A depicts this for all and cash-only transfer programs in total, and Panel B shows the results by program. We explain the construction of replicating tax changes in the text. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands.

distribution is 13.9 percent for all transfers together versus 24.7 percent for the replicating income tax. This pattern applies broadly across programs, with the exception of UI.

## **6 Conclusion**

Distributional analyses that rank households by their current incomes are known to overstate the progressivity of income taxes and the regressivity of consumption and excise taxes. This paper presents a further criticism of such analyses: They obscure the particular progressivity of transfer programs in consumption and lifetime terms. Unlike typical taxes, transfer programs reduce inequality in consumption and lifetime income by about as much as they reduce inequality in annual income.

Among low-income households, transfer programs detect (through eligibility rules) and attract (through selective take-up) those whose low incomes are not transitory and thus who cannot support higher consumption through smoothing over time. Most of this targeting comes from selection via take-up rather than selection via eligibility. Insofar as income taxes cannot target via selective take-up, shifting the burden of redistribution from transfer programs to income taxes—such as through a universal basic income—would require substantially higher marginal tax rates than is widely understood if this shift is not to increase consumption poverty.

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